# PROJECT REQUIREMENTS

Enterprise SIEM Lab: Wazuh Cluster, Log Aggregation, and Custom Alert Dashboards

## 1. Hardware Requirements

• Development workstation: 4 CPU cores and 8 GB RAM recommended for working with exported alert data.
• SIEM lab: resource needs depend on event volume and deployment mode; a multi-node cluster requires additional CPU, memory, and storage.
• Sufficient disk capacity for retained logs and indexes.

## 2. Software Requirements

• Python 3.10 or newer for the companion reporting utility.
• Wazuh manager, indexer, and dashboard for the full SIEM lab, installed using official version-specific documentation.
• Wazuh agents on approved Windows/Linux test endpoints, where needed.
• Browser and terminal access; optional virtualization platform and isolated virtual network.

## 3. Python Dependencies

No third-party Python packages are required for the included reporting utility. It uses the Python standard library for JSON parsing, CSV generation, counters, timestamps, and command-line options.

## 4. Functional Requirements

• Read Wazuh alert exports as JSON arrays/objects or JSON Lines.
• Normalize timestamp, agent, rule, level, description, location, and message fields.
• Summarize alert counts by severity, rule, and agent.
• Filter records by minimum rule level.
• Generate dashboard-friendly JSON and a CSV triage queue.

## 5. Operational and Security Requirements

• Collect logs only from authorized endpoints and systems.
• Use an isolated lab network for controlled exercises.
• Sanitize exported logs before sharing or committing them.
• Restrict dashboard and API access with strong authentication and least privilege.
• Define log retention, time synchronization, backup, and alert-review procedures.

## 6. Acceptance Criteria

• The sample input processes successfully and creates valid JSON and CSV outputs.
• A minimum-level filter includes only matching or higher-level events.
• Summary counts and normalized fields are traceable to the input records.
• Documentation distinguishes the file-processing utility from actual Wazuh cluster deployment.

Note: Exact cluster sizing, installation steps, and supported operating systems should be verified against the official Wazuh documentation for the version selected.